QUANT TRADING & ALPHA EXPLAINED — plus the one-cell offline+online research design

Session 26 · 2026-10-27 · owner-requested deep explanation (web + paper researched) Covers: what quant trading is · how the math works · how alpha is found · what finding it needs · the per-symbol research cell (offline) and how online cells consume proven artifacts · one-cell two-mode design · candlestick "sighting" · Kronos foundation-model evaluation. LLMs never in hot path (R2/R3). Every claim is measured, web-verified with source, or marked unverified.

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§1 · What quant trading IS

	Discretionary trader	Quant trader
Decision from	experience, gut, charts	**a mathematical rule computed from data**
Can explain the trade?	"I feel the trend"	"signal X > threshold, because historically this predicted Y with IC 0.04"
Speed	minutes–hours of thinking	microseconds–seconds of code
Consistency	changes mood to mood	identical in every condition
Scaling	1 human = 1 chart	1 system = 73 symbols

One sentence: quant trading = *turning a statistical edge into a repeatable rule, then letting computers execute it without emotion.*

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§2 · How it works — the pipeline

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DATA → SIGNAL (alpha) → PORTFOLIO → EXECUTION → RISK → LEDGER
        ▲                                              │
        └──────────── feedback (what actually happened) ──┘
```

Stage	What happens	Our engine
1. Data	pull prices, news, fundamentals, book — stamped "as-of" time	E1 + data layer
2. **Signal (alpha)**	compute a score predicting future return	E1 features → E3 model
3. Portfolio	convert score → position size	allocator
4. Execution	place orders with minimal cost (TWAP/VWAP/limits)	E6
5. Risk	cap loss, kill switch, drawdown ladder	E5 (deterministic)
6. Ledger	record everything, reconcile nightly	E7
7. Feedback	measured P&L → retrain or retire the signal	E3/E4

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§3 · The math

Alpha (Jensen) — what "alpha" literally means

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Your return = β × market return  +  α
              (riding the market)     (your actual edge)
```

`α = R_portfolio − [R_f + β(R_market − R_f)]`

If β explains all the profit you just held an index and paid fees. **Alpha = what remains after removing everything explainable.**

The Fundamental Law — Grinold & Kahn (1989/2000)

`IR ≈ IC × TC × √BR`

Term	Meaning	Real-world value
IR	information ratio = active return per unit risk	> 1 good, > 1.5 excellent
IC	forecast-vs-outcome correlation	**0.02–0.07 real markets; > 0.10 suspicious = usually overfitting**
TC	transfer coefficient — signal→trade quality after costs	0.7–0.9 realistic
BR	breadth = independent bets per year	the multiplier

Worked numbers:

Scenario	IC	BR	TC	IR	Verdict
One symbol, 1 bet/month	0.05	12	0.9	**0.16**	❌ useless
One symbol, 1 bet/day	0.05	250	0.9	**0.71**	⚠️ okay
One symbol, ~10 independent bets/day (scalping)	0.03	2,500	0.7	**≈ 1.05**	✅ good
600 stocks monthly (institutional)	0.05	72,000	0.8	~10 theoretical	the fund game

Three proofs from this law: (1) small IC is normal — 0.05 ≈ 52.5% pair-ranking, edge comes from repetition; (2) one symbol works *only* at high bet frequency; (3) bets must be **independent** — correlated bets collapse √BR.

Honesty formula — Deflated Sharpe Ratio (Bailey & López de Prado 2014)

```
Test 1,000 variants → the BEST looks great by luck.
DSR = probability your Sharpe is real after counting every trial run.
```

Our gate DSR ≥ 1. The research cell must log `trials_run`; DSR is computed against total search count. *This is the single most important line of code in the research agent.*

Decay formula — why speed matters

Finding	Source
Published anomalies lose **~50%** of returns post-publication	McLean & Pontiff 2016
After costs, decay reaches **~93%** for the typical anomaly	Cambridge JFQ 2023
Decay worsens **~5 ppt per year**	*Quantitative Finance* 2022
Only **7.58%** of stock alpha estimates are statistically significant (t>2)	BFI Chicago WP 2024

Private/unpublished alpha decays slower — **finding your own beats copying papers.**

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§4 · How alpha is found — 5 methods

#	Method	How	Non-LLM?	Evidence
1	**Hypothesis-first**	economic logic → test	✅	the classical way
2	**Formula search (GP / symbolic regression)**	machines compose/mutate formulas 100,000×, score by IC	✅	AlphaForge (AAAI 2025), RiskMiner (MCTS), warm-start GP (arXiv 2412.00896), gplearn
3	**ML pattern discovery**	GBM/NN learn mapping	✅	our E3
4	**Alternative data**	satellite, card, news NLP	✅ (NLP separate)	AIMA: leaders' top choice
5	**Microstructure**	order book (OFI, microprice)	✅	FX scalping plan §3

Methods 1+2+3 are what the non-LLM research agent runs. GP = "build your own alpha formulas automatically" — worked since the 1990s, no LLM needed.

§5 · What you NEED to find alpha — the six

#	Requirement	Do we have it?
1	Point-in-time data (no leakage)	✅ PIT assembler
2	Compute (search 10^4–10^6 formulas)	✅ E3 + Colab GPU
3	Statistical discipline (DSR, CPCV, trial logging)	✅ E4 gates
4	Breadth (√BR)	✅ 73 cells = portfolio BR
5	Cost model in the label	✅ E7 feedback
6	Speed + privacy (decay clock)	⚠️ the real bottleneck

§6 · The per-symbol RESEARCH CELL (offline) — spec

The owner's ask: give the offline agent hypothesis-first + GP/symbolic search + ML discovery + alternative data + microstructure + fundamentals/technicals/indicators/strategies/candlesticks — and have online agents use what proves out.

Owner's ingredient	Research-cell implementation	Status
Hypothesis-first	economic-logic templates → candidate statements, each with a falsification test	to build
Formula search (GP/symbolic)	expression-tree search over operators (rank, corr, decay, rolling, OHLC math), fitness = IC, **every trial logged**	to build (proven method)
ML pattern discovery	E3 GBM/small-NN training on candidate features	✅ engine exists
Alternative data	FRED vintages, BLS, CFTC COT, FXStreet, news as-of	✅ sources named (FX plan §3)
Microstructure	OFI, microprice, book imbalance, spread regime	✅ spec'd (FX plan §3)
Fundamentals/technicals/indicators	standard E1 feature library (RSI, ATR, MA, momentum, carry, …)	✅ standard
Candlestick patterns	pattern detectors as **binary features** (body/range, engulfing flags, trend structure)	features ✅ — evidence mixed, never hard signals
Outputs	candidates + evidence pack (trials_run, IC, DSR, crisis replay, net-of-cost)	gate → E4

§7 · How ONLINE agents use what offline proved — the handoff

The offline agent does not "teach" the online agent. It produces **frozen artifacts**; the online agent executes them without thinking:

```
OFFLINE RESEARCH CELL                  E4 ARENA (external gate)        ONLINE TRADING CELL
search 10,000s of candidates ──────► CPCV + DSR≥1 + crisis ──────► load frozen artifact
log EVERY trial (DSR)                replay · shadow 3–6 mo            compute each tick
                                     (cell CANNOT edit this)           NEVER searches
```

What research proves	Artifact	How the online agent uses it
a formula works	**expression + params** in R3 registry	E1 evaluates every bar/tick
an ML model works	**ONNX file** in R3	E3 deterministic inference
a threshold works	**JSON config** (e.g., "no entry when spread > 1.5 pip")	E1/E5 enforce each cycle
a feature set works	**feature spec** (columns + as-of rules)	E1 builds exactly those
a style works for the symbol	**spec-sheet update** (one style)	trading cell runs that style
edge decaying	**decay monitor** (rolling IC watch)	E7 → E4 demotes to research

Core rule: *online = execute frozen artifacts only. Research = create candidates only. Neither crosses without E4.*

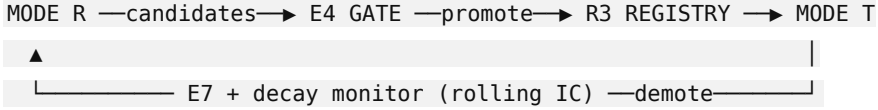
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§8 · ONE cell, offline + online, one symbol — ONE CELL · TWO MODES · EXTERNAL GATE

Session 22 rejected *one agent trading all styles simultaneously*. This design is different: **sequential modes on one symbol**, gate outside the cell.

	MODE R (research, offline)	MODE T (trading, online)
Runs	batch, hours/days	continuous, µs–ms
May fail	**yes** — failure is data	**no** — governed
Searches?	✅ all 6 ingredient groups	❌ registry reads only
Places orders?	❌ **no E6 access**	✅ promoted artifacts only
Edits itself?	candidates only	**never**
LLM allowed?	optional (R3 permits research)	**never** (R2)
Output	candidates + trial log	P&L + ledger

State machine:



Guard rules:

#	Rule	Defeats
G1	**E4/E5 are separate components the cell cannot write**	self-grading
G2	**Mode T has no search code path** (not loaded)	hot-path contamination
G3	**Mode R has no order code path**	research placing trades

#	Rule	Defeats
G4	**Every trial logged → DSR against total search count**	overfitting
G5	**One style active in Mode T at a time** (Session 22 preserved)	negative transfer
G6	**Mode R and Mode T run as separate processes** sharing the registry	research crash must not stop live trading

Performance (honest): research quality = identical to separate agents (same search, 73 cells parallel); trading quality = identical **if G1–G5 hold**; main risk = gate integrity; no conflict with Session 22 (that evidence bans multi-style *simultaneous* trading weights, not research-then-trade sequence).

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§9 · Can the agent "see" candlesticks like us?

Question	Answer
Eyes?	**No** — it receives OHLCV as numbers
Worse than seeing?	**No — better.** The chart is a visual compression of those numbers; the agent gets the raw thing humans interpret
Chart images (vision model)?	possible, no proven edge over numeric, slower, harder to audit → **not recommended**
"Reading candles" for it	pattern features: body/range ratio, higher-high structure, engulfing flags, regime
Kronos?	**tokenizes OHLCV into discrete tokens and reads them like a sentence** — machine-form "seeing candles" from numbers, not images

Candlestick-pattern evidence (mixed — features, not signals):

Study	Market	Result
Marshall & Young 1998	S&P 500	✅ 36σ, ~1%/2-day
Chen/Bao/Zhou 2016	China	✅ 5/8 patterns significant (mid/large caps)
Lu 2012	Taiwan electronics	✅ 3 bullish patterns profitable
Tharavanij 2017	Thailand	❌ most patterns not ≠ 0
Brazil 2013	Ibovespa	❌ few confirmed; one significant *opposite*
Vietnam 2018	Vietnam	❌ none of 10 significant
Engulfing 2021	S&P 500	⚠️ some outperformance

Verdict: legitimate features, unreliable standalone signals — market-dependent. Feature bank yes; hard-code "engulfing = buy" no.

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§10 · Kronos (shiyu-coder/Kronos) — verified facts vs independent evidence

What it is (verified)

Fact	Value
Repo	`shiyu-coder/Kronos` — **29,561 stars**, **MIT**
What	decoder-only foundation model for **K-line (candlestick) sequences**
Training	**12 billion K-line records, 45 global exchanges**

Fact	Value
Mechanism	tokenizer: OHLCV → hierarchical discrete tokens → autoregressive prediction
Paper	arXiv 2508.02739 · **accepted AAAI 2026**
Claim	**RankIC +93%** vs leading TSFM, +87% vs best non-pretrained; vol MAE −9%; generative fidelity +22%
Sizes	Kronos-**mini = 4M params** (demo: 360-bar context, 30 MC paths → probability distribution of future candles)
Extras	fine-tuning scripts (2025-08)

Independent evaluation (the part the abstract doesn't show)

Independent study	Protocol	Result
`kronos-market-eval` (2026-08-04)	walk-forward, BTC + SPY, vs random walk	❌ **"Kronos does not beat 'the price stays where it is'"**
Finance Research Letters 2026	**4,998 A-shares, lagged-only OOS, 2023–2025**	❌ zero-shot MAE **2.4–2.6× worse than HAR-RV**, **last place every year**; fine-tuning **collapses encoder**; vs GARCH hybrid d=0.06 (p=0.42) = no difference
Diagnosis	—	BSQ quantization + price-derived volatility explain failure

Verdict

Question	Answer
Add it?	✅ **as a Mode-R CANDIDATE, not an oracle**
Role	**feature generator**: forecast distribution → features (`P(up)`, expected range, forecast vol, 30-path spread) into E1/E3
LLM ban?	**Not an LLM** — 4M-param TSFM = E3-class model. Allowed offline; online only as frozen artifact
Gates	CPCV + DSR + shadow + decay monitor — **default expectation = fails** (two independent negative results)
Fine-tune on 73 symbols?	MIT allows; encoder-collapse evidence weak → zero-shot baseline first
Hot path (scalping)?	⚠️ latency unverified — autoregressive + 30 MC paths likely too slow for ms-class; **fine for H1/D1 swing cells**
One-liner	*Promising candle-language reader; published wins not yet reproduced by outsiders — feature, not forecast god.*

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§11 · Bottom line

Question	Answer
What is quant trading?	Statistical edge → repeatable rule → computer executes without emotion
How does the math work?	**α = return unexplained by beta · IR = IC × TC × √BR · DSR = honesty · decay ≈ 50%+**
How is alpha found?	hypothesis · **GP formula search (non-LLM, proven)** · ML · alt data · microstructure
What's needed?	PIT data + compute + DSR discipline + breadth + net-cost labels + speed
One non-LLM research agent per symbol?	✅ build it — formalized Invention Lane
Same agent also trading everything?	❌ self-grading, negative transfer, hot-path rules → **one cell, two modes, external gate (G1–G6)**
How do online cells use proven work?	**frozen artifacts** via R3 — execute, never search

Question	Answer
Can it see candles?	better than us — raw OHLCV; patterns = features (evidence mixed)
Add Kronos?	✅ Mode-R candidate feature generator — but independent evals say it doesn't beat naive baselines → must clear our gates assuming failure

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§12 · Provenance

Claim	Class
IR = IC × TC × √BR; IC norms 0.02–0.07	**Web-verified** — Grinold-Kahn, CFI, QuanterLab, riskhub
DSR / backtest overfitting	**Web-verified** — Bailey & López de Prado 2014, AMS 2014
Alpha decay 50% / 93% / +5ppt-per-year / 7.58% significant	**Web-verified** — McLean-Pontiff 2016, Cambridge JFQ 2023, Quant Finance 2022, BFI Chicago 2024
GP/symbolic alpha mining works	**Web-verified** — AlphaForge AAAI 2025, RiskMiner, arXiv 2412.00896
Kronos facts (MIT, 12B records, AAAI 2026, +93% RankIC)	**Web-verified** — GitHub + arXiv 2508.02739
Kronos independent failures	**Web-verified** — kronos-market-eval 2026-08-04, Finance Research Letters 2026
Candlestick pattern results	**Web-verified** — 7 studies in §9, mixed
One-cell-two-mode design	**Design proposal** — guard rules G1–G6 are requirements, not measurements
Research-cell performance	**Unproven until built** — identical-to-split is conditional on gate integrity

Artifact of Session 26. Companion to FX-SCALPING-BUILD-PLAN.md (Session 25) and the 73-cell first batch (Session 24).